



Zig API Test Interface

✓ Server is online and responding



Test All Endpoints

API Endpoints

GET http://localhost:8080/components/

Test

Description: Get all components from the database

✓ Success (200) - 3ms

```
[
  {
    "id": 1,
    "type_id": 1,
    "value": "10kΩ",
    "quantity": 500,
    "footprint": "0805",
    "vendor_id": 1,
    "description": "Surface mount resistor 10kΩ ±1%",
    "vendor_part_number": "RC0805FR-0710KL",
    "price": "0.02"
  },
  {
    "id": 2,
    "type_id": 2,
    "value": "100nF",
    "quantity": 1000,
    "footprint": "0603",
    "vendor_id": 2,
    "description": "Ceramic capacitor 100nF 50V X7R",
    "vendor_part_number": "C0603X7R500-104K",
    "price": "0.05"
  },
  {
    "id": 3,
    "type_id": 3,
    "value": "10μH",
    "quantity": 200,
```

```
    "footprint": "SMD 1210",
    "vendor_id": 3,
    "description": "Shielded power inductor 10µH ±20%",
    "vendor_part_number": "SD1210-100K",
    "price": "0.3"
  },
  {
    "id": 4,
    "type_id": 4,
    "value": "1N4148",
    "quantity": 750,
    "footprint": "D0-35",
    "vendor_id": 4,
    "description": "Small signal switching diode",
    "vendor_part_number": "1N4148-T",
    "price": "0.01"
  },
  {
    "id": 5,
    "type_id": 5,
    "value": "2N3904",
    "quantity": 300,
    "footprint": "T0-92",
    "vendor_id": 5,
    "description": "NPN general purpose transistor",
    "vendor_part_number": "2N3904BU",
    "price": "0.04"
  }
]
```

GET http://localhost:8080/components/:id

Test

Description: Get a specific component by its ID

Component

ID:

1

✓ Success (200) - 2ms

```
{
  "id": 1,
  "type_id": 1,
  "value": "10kΩ",
  "quantity": 500,
  "footprint": "0805",
  "vendor_id": 1,
  "description": "Surface mount resistor 10kΩ ±1%",
  "vendor_part_number": "RC0805FR-0710KL",
  "price": "0.02"
}
```

GET http://localhost:8080/vendors/

Test

Description: Get all vendors from the database **Success (200) - 3ms**

```
[
  {
    "id": 1,
    "url": "https://www.digikey.com",
    "description": "Electronics components distributor",
    "name": "DigiKey"
  },
  {
    "id": 2,
    "url": "https://www.mouser.com",
    "description": "Global distributor of semiconductors and components",
    "name": "Mouser"
  },
  {
    "id": 3,
    "url": "https://www.arrow.com",
    "description": "Electronic components and enterprise computing",
    "name": "Arrow Electronics"
  },
  {
    "id": 4,
    "url": "https://www.tme.eu",
    "description": "European electronic components supplier",
    "name": "TME"
  },
  {
    "id": 5,
    "url": "https://www.sparkfun.com",
    "description": "Hobbyist electronics and prototyping parts",
    "name": "SparkFun"
  }
]
```

GET http://localhost:8080/vendors/:id

Test

Description: Get a specific vendor by its ID**Vendor ID:**  **Success (200) - 2ms**

```
{
  "id": 1,
  "url": "https://www.digikey.com",
  "description": "Electronics components distributor",
```

```
"name": "DigiKey"
}
```

GET http://localhost:8080/types/

Test

Description: Get all types from the database

✓ Success (200) - 2ms

```
[
  {
    "id": 1,
    "name": "Resistor",
    "description": "Passive component used to limit current flow"
  },
  {
    "id": 2,
    "name": "Capacitor",
    "description": "Passive component used to store electrical charge"
  },
  {
    "id": 3,
    "name": "Inductor",
    "description": "Passive component used to store energy in a magnetic field"
  },
  {
    "id": 4,
    "name": "Diode",
    "description": "Component that allows current to flow in one direction"
  },
  {
    "id": 5,
    "name": "Transistor",
    "description": "Active component used for switching and amplification"
  }
]
```

GET http://localhost:8080/types/:id

Test

Description: Get a specific type by its ID

Type ID:

✓ Success (200) - 3ms

```
{
```

```
    "id": 1,  
    "name": "Resistor",  
    "description": "Passive component used to limit current flow"  
  }
```

Data Structures

Component Data Structure

The Components endpoint returns objects with the following structure:

```
{  
  "id": 1,  
  "type_id": 1,  
  "value": "component_value",  
  "quantity": 100,  
  "footprint": "component_footprint",  
  "vendor_id": 1,  
  "description": "Component description",  
  "vendor_part_number": "VPN123",  
  "price": "9.99"  
}
```

Vendor Data Structure

The Vendors endpoint returns objects with the following structure:

```
{  
  "id": 1,  
  "name": "vendor_name",  
  "description": "Vendor description",  
  "url": "https://vendor.com"  
}
```

Type Data Structure

The Types endpoint returns objects with the following structure:

```
{  
  "id": 1,  
  "name": "type_name",  
  "description": "Type description"  
}
```

